

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We examine how physical scale, population size, and sensory capacity shape behavioral emergence in open-ended ecological simulations. Agents receive no explicit rewards, demonstrations, or task objectives; instead, populations persist through resource acquisition, reproduction, mutation, and selection while continuously altering both the environment and the surrounding population. The objective of the study is therefore not to optimize a single policy for a predefined task, but to characterize the ecological conditions under which structured behavioral repertoires arise in large evolving populations. Using modern accelerators and a new JAX-based multi-agent simulator, we scale objective-free ecological evolution to worlds containing more than 1,000,000 grid cells and populations exceeding 60,000 agents, each controlled by its own evolved neural network policy. Across these settings, we document several reproducible forms of adaptive ecological behavior, including long-range resource extraction, vision-guided foraging, and predation. The experiments show that these behaviors depend systematically on both sensory information and scale: compass information enables reliable long-range resource transfer in directionally structured terrains, vision improves foraging and attack precision in ocean environments, and larger worlds make several population-level dynamics more stable across runs. These results do not establish open-ended intelligence in the unrestricted sense; rather, they demonstrate that large, resource-coupled ecological simulations provide a practical experimental substrate for studying how survival, competition, and population turnover can generate nontrivial artificial behaviors without explicit task rewards. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been a central object of inquiry in evolutionary theory (Darwin, 1859; Simpson, 1944), ecology (MacArthur & Wilson, 2001), and genetics (Dobzhansky, 1937). Models of these dynamics range from theories of speciation and adaptive radiation (Coyne & Orr, 2004; Schluter, 2000) to the mathematical foundations of population genetics (Fisher, 1930; Wright, 1931).

Mathematical analysis, laboratory experimentation, and field observation have clarified many mechanisms by which evolution operates in natural systems. Yet the ecological richness that makes these systems scientifically important also makes them difficult experimental objects: large heterogeneous ecosystems are hard to control, difficult to measure exhaustively, and ethically costly to perturb. Interventions at the spatial and temporal scales relevant to open-ended adaptation may damage, displace, or otherwise alter the very populations whose dynamics they are intended to reveal.

We address this constraint by studying the open-ended evolution of neural network policies in large-scale ecological simulations. Agents in our setting receive no externally specified reward or task objective. Instead, they collect resources, survive, reproduce, and mutate according to the dynamics of the environment, with policy variation introduced when parents transmit mutated policies to offspring. This design turns the simulator into an experimental system for asking how competitive and environmental pressures, rather than externally imposed rewards, generate behavioral structure in populations of digital organisms. With modern accelerator hardware, we can examine these dy-

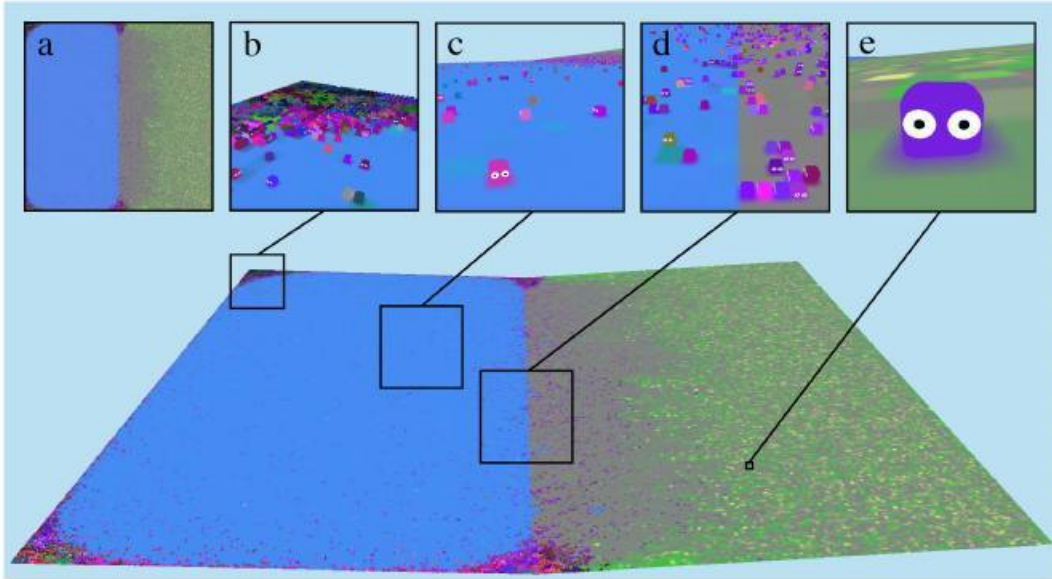


Figure 1: A 3D visualization of a 512×512 map in our environment. Inset (a) shows a top-down view of the complete map; (b) shows agents clustered in a corner; (c) shows agents in water; (d) shows agents moving between land and water; and (e) shows an individual agent gathering local resources.

namics at scales where environmental extent and population size become experimentally visible variables rather than fixed background assumptions.

To support these experiments, we use a new JAX-based simulation environment for high-throughput evaluation of large ecological grid worlds. Our largest experiments contain more than 60,000 individual agents distributed over more than 1,000,000 grid cells. Agents navigate to obtain water, energy, and biomass; agents that accumulate sufficient resources reproduce by creating mutated copies of themselves. Figure 1 illustrates the resulting environment in 3D.

The broader motivation is to study ecology not only as an object of simulation, but also as a mechanism for producing artificial behavior. Recent work in AI has shown that scale can do more than improve performance; beyond particular thresholds, it can qualitatively transform the capabilities of learned systems (Wei et al., 2022a). Nevertheless, many embodied learning studies remain limited to comparatively small worlds, fixed tasks, and modest populations. Our work extends the scaling question from individual models to entire evolving ecosystems. Analogously to the way reasoning abilities may appear in individual models only beyond particular scale regimes (Wei et al., 2022b), we examine which population-level strategies arise only when environmental size, population count, and ecological complexity become sufficiently rich.

The paper makes three contributions. First, we introduce and instantiate an objective-free ecological game framework in which populations grow and shrink through endogenous birth, death, resource consumption, and mutation rather than through a fixed reinforcement-learning objective. Second, we show that sensory structure can determine ecological outcomes: compass-equipped agents develop long-range resource-gathering behavior in directionally structured terrains, whereas otherwise comparable agents without compass information often fail to recover biomass deposited on land. Third, we demonstrate that scale affects not only average population outcomes but also the reliability and stability of emergent behaviors, and that vision enables more effective foraging and more precise predation in the **Ocean** environment, with clearer advantages at larger grid sizes. Taken together, these findings identify ecological scale and sensory richness as measurable drivers of behavioral emergence in large evolving populations, while delimiting the claim to the resource-coupled environments and event definitions studied here.

2 RELATED WORK

Computational studies of artificial life have long used simple mechanistic rules to investigate how organized population-level behavior can arise. Conway’s Game of Life (Games, 1970) demonstrated that local cellular updates can generate complex spatial patterns, while Tierra (Ray, 1992) and Avida (Brown) modeled digital life as self-replicating programs subject to mutation. Relatedly, Reynolds (1987) showed that local motion rules are sufficient to produce flocking and schooling behavior. These systems established the central methodological premise that rich collective dynamics can emerge from compact local rules, but they were developed under computational regimes far smaller than those now available.

Evolutionary computation has also been used to design agent bodies, sensors, and policies. Prior studies have evolved locomotor morphology (Sims, 1994a, b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing mechanisms (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025), while neuroevolution methods have optimized neural architectures directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). Our emphasis differs from this design-oriented tradition. Rather than using evolution as an optimizer for a specified performance criterion, we use evolutionary mechanisms to study which behaviors are selected by resource constraints, spatial structure, and population turnover in objective-free ecological environments.

Evolution has likewise supplied the internal logic of many optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) formalize selection and mutation as search procedures, and related methods encourage novelty in the resulting solutions (Lehman & Stanley, 2011; Cully et al., 2015). These techniques have been applied to neural network optimization for Atari games and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). Although our agents are updated by mutation and selection, the purpose of the present work is not to introduce a new optimizer or to maximize a fixed benchmark reward. We instead treat evolution as a population-level process whose behavioral consequences can be measured under controlled ecological conditions.

A closely related line of research uses biological inspiration to place organisms inside simulated environments. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). More recent work uses cellular automata to produce organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025). Other simulated environments have been used to study emergent language and communication (Mordatch & Abbeel, 2018; Park et al., 2023). Autocurricula, in which interacting agents create an increasingly challenging learning landscape for one another (Leibo et al., 2019), have also been studied in populations smaller than those considered here (Leibo et al., 2018; Team et al., 2021). Conversely, Wang et al. (2019) and Aki et al. (2024) evolve environments designed to improve agent learning. Although these works differ in whether the organism, policy, or environment is the primary object of adaptation, they collectively motivate our focus on environmental complexity as a driver of emergent behavior.

The most directly comparable studies simulate populations of agents interacting in complex worlds. Baker et al. (2019) examine a Hide-and-Seek task in which a small number of reinforcement-learning agents acquire tool-use strategies. Other multi-agent systems train populations for specified objectives, including video-game play (Jaderberg et al., 2019), object manipulation (Petrenko et al., 2021), and capture tasks (Zheng et al., 2017; Yamada et al., 2020). These systems demonstrate compelling forms of emergence, but they generally retain fixed tasks, reward functions, or population sizes. Neural MMO (Suarez et al., 2019, 2023) moves toward more open-ended reinforcement learning by optimizing agents for survival in a massively multi-agent game. However, its experiments cap the number of concurrent agents at 128 and do not include generational inheritance, which limits the study of endogenous demographic dynamics. Hamon et al. (2023) is closer to our setting through variable population size, genetically inspired evolution, and generational timelines, but uses a substantially smaller and less feature-rich environment and does not directly isolate scale. Similarly, Lu et al. (2024) simulates agents evolving to gather resources, but operates at a maximum of 256 agents and emphasizes tool use through programmable robots. Our work combines endogenous population change, mutation-based inheritance, rich resource and terrain dynamics, and

substantially larger populations, enabling empirical questions about scale and ecological stability that are difficult to address in these prior systems.

Finally, we build on recent progress in accelerated simulation. Optimized JAX and GPU-native environments have increased the speed and scale of game-like learning systems (Koyamada et al. 2023; Matthews et al. 2024; Lechner et al. 2023). We apply similar acceleration principles to ecological evolution, where the computational challenge is not only stepping many agents quickly, but also supporting births, deaths, mutable neural policies, resource redistribution, and large spatial worlds over millions of steps.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize an evolving population and its environment as an **Ecological Game (EG)**, a multi-player dynamical system for objective-free evolutionary settings. As in Markov decision processes (MDPs) and stochastic games (SGs) (see Sutton et al. (1998) and Albrecht et al. (2024)), an EG has Markovian transition dynamics, a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function T that maps a state s_t and the agents' actions $a_{t,i}$ to a distribution over successor states s_{t+1} . Unlike an SG, an EG omits an externally specified reward function and instead includes a population function $\Gamma(s_t)$, which identifies the agents in I that are alive in state s_t and records parentage so that births and deaths are represented as part of the state evolution. As with partially observable variants of MDPs and SGs, an EG may restrict state access through observations; we call this setting a POEG. A POEG replaces full-state access with an observation function $O(s_t, i)$ that maps the current state and an agent identity to the agent's observation. The environment used in this paper is a POEG whose observation function is determined by the sensor configurations described below.

3.2 ENVIRONMENT

The POEG used in our experiments is a large grid world in which agents must acquire resources to remain viable and to reproduce. The environment state is factored as $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, where s_t^m contains map-level variables, including terrain and resource distributions, and s_t^a contains agent-level variables such as position, health, and internal resource stores.

Terrain: Each map contains a rock layer specifying the elevation of every grid cell. Water is initially assigned to cells whose rock value lies below a preset sea level and then evolves during the episode according to a discrete downhill-flow simulation induced by the rock layer. The experiments use the six global terrain configurations shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. We evaluate grid sizes of 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 . When grid size changes, the initial population is scaled in proportion to map area; after initialization, population size is determined endogenously by the environmental dynamics.

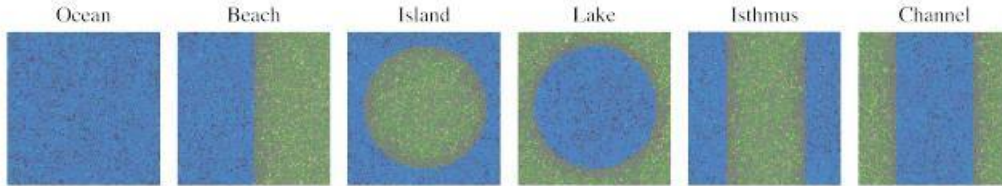


Figure 2: Overhead views of the terrains used in our experiments at 256×256 grid size. Small dark spots denote individual agents, while green and yellow regions indicate concentrations of energy and biomass. Each image was captured 5000 steps into an episode, after most water biomass had been consumed but before agents had evolved longer-term survival on land.

Resources and Agent Health: Agents require **water**, **energy**, and **biomass** for survival and reproduction. At initialization, biomass and energy are randomly distributed across the map; during an episode, agents consume, carry, and redistribute these resources through their actions and deaths.

Every action consumes a small amount of energy and water. Spent water returns to the landscape and flows downhill toward the nearest body of water, whereas spent energy is removed from the system. Biomass determines reproduction: each agent requires a fixed amount of biomass to exist and can reproduce only after accumulating sufficient excess biomass to allocate to an offspring. Total biomass, like total water, remains conserved over the simulation, but biomass does not flow autonomously. Energy is not conserved; instead, it slowly grows on free-standing biomass. These coupled resource rules create an ecological control problem that is richer than collecting a single undifferentiated food source. For ablations on resource rules, see Appendix F.2

Agents also maintain health points (HP), which determine survival. If an agent overspends a critical resource, its HP deteriorates rapidly, although agents can expend energy and water to recover HP gradually. Agents may also attack other agents, reducing the targets' HP, and each agent incurs a small age-dependent damage term that increases over time, ensuring eventual death from old age in the absence of other causes. When an agent's HP reaches zero, the resources it carries are deposited into its current grid cell. Consequently, an agent that kills another agent may subsequently consume the victim's deposited resources, producing a simple predatory interaction. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Agents observe the environment through four possible sensors. The **internal** sensor reports the agent's age, health, and carried resources; the **external** sensor reports resources in the agent's current grid cell; the **compass**, motivated by magnetoreception in birds and other animals, encodes global orientation as a four-way one-hot vector; and **vision** supplies a 7×7 top-down rendered patch centered on the agent, with three color channels and a fourth channel representing local elevation difference. For ablations on vision configurations, see Appendix F.6

Actions: Agents choose from a finite discrete action set organized into five categories. **Rest** performs no environmental operation and has the lowest resource cost. The three **Move** actions rotate the agent clockwise, rotate it counterclockwise, or move it forward by one grid cell. Cell occupancy is exclusive: if an agent attempts to enter an occupied cell, or if multiple agents attempt to enter the same cell, the corresponding move action is canceled. The **Attack** action kills all agents in the 3×3 square directly in front of the acting agent. For ablations on attack strength, see Appendix F.7 The three **Eat** actions correspond to the three resource types. If the agent occupies a cell containing the selected resource, the action transfers a fixed amount of that resource from the environment into the agent's finite-capacity stomach.

Finally, **Reproduce** implements single-parent reproduction. The action succeeds only when the parent has accumulated enough resources to allocate to a new child. The child is created directly behind the parent, receives a mutated copy of the parent's policy weights, and immediately inherits a fixed quantity of the parent's resources.

To isolate the ecological consequences of different capabilities, we selectively disable sensors and the attack action across experiments. We use three sensing configurations: internal and external resource sensors only (**R**), internal and external sensors plus compass (**RC**), and internal and external sensors plus compass and vision (**RCV**). The suffix **+A** denotes that attacking is enabled; for example, (**RC+A**) denotes agents with resource sensors, a compass, and the attack action.

3.3 POLICIES

Agent policies are small memoryless MLPs. We deliberately omit recurrent memory to keep the computational model simple and to reduce per-agent parameter count. Raw sensor values are flattened, normalized to $[-1, 1]$, and passed through separate linear layers with output dimension 64. The resulting embeddings are summed and processed by a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer is linear with output dimension equal to the number of available actions. Actions are sampled from a softmax distribution whose temperature evolves with the network weights. We represent weights using half-precision bfloat16 values to support larger populations and faster runtimes. Depending on the active sensors, each agent network contains roughly 10k–25k parameters. For ablations on network size and architecture, see Appendix F.5

For every run, the initial agents' weights are sampled using Kaiming initialization (He et al., 2015), and bias vectors are initialized to zero. Policies for agents born during an episode are generated

by copying the parent’s weights and adding Gaussian noise with standard deviation 3×10^{-2} to all mutable weight values. This mutation scheme follows the simple perturbation-based structure used in neuroevolution work such as (Such et al., 2017). Although more elaborate mechanisms such as crossover are possible, we use this minimal reproduction rule to focus the empirical analysis on environmental scale and sensory capability. Empirically, low-precision floating-point parameters required a larger mutation rate. Appendix F.4 reports the corresponding mutation-rate ablation.

Each agent also has a three-channel trait that determines its rendered color. This color trait mutates analogously to the network weights and specifies how the agent appears to other agents with vision. In principle, this channel allows visually equipped agents to distinguish among different visually rendered agent types.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al., 2018) and executed on Nvidia H100 and A40 GPUs using a high-throughput simulation environment that computes environment dynamics, agent updates, and measurements on accelerator hardware. Unless otherwise specified, each run proceeds for 2M environment steps or until population extinction. Per-seed wall-clock time ranges from under one hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on a single H100 GPU. Appendix E provides additional simulator details, scaling measurements, and reproducibility information.

4 EXPERIMENTS

The experiments evaluate how sensory modalities, environmental scale, and population size affect the behaviors selected by the ecological dynamics. We focus on two representative phenomena: long-distance resource gathering in terrains with structured land–water geometry, and vision-mediated foraging and predation in the **Ocean** environment.

4.1 LONG-DISTANCE RESOURCE GATHERING

The first experiment asks whether agents can evolve policies that travel inland to recover resources and then return to water. In terrains with clear directional structure, agents equipped with a compass (**RC**) develop substantially more reliable long-range resource-gathering behavior than the baseline agents (**R**) that have resource sensors but no compass. The same experiment also tests scale: the behavior appears more consistently and supports more stable populations in larger environments.

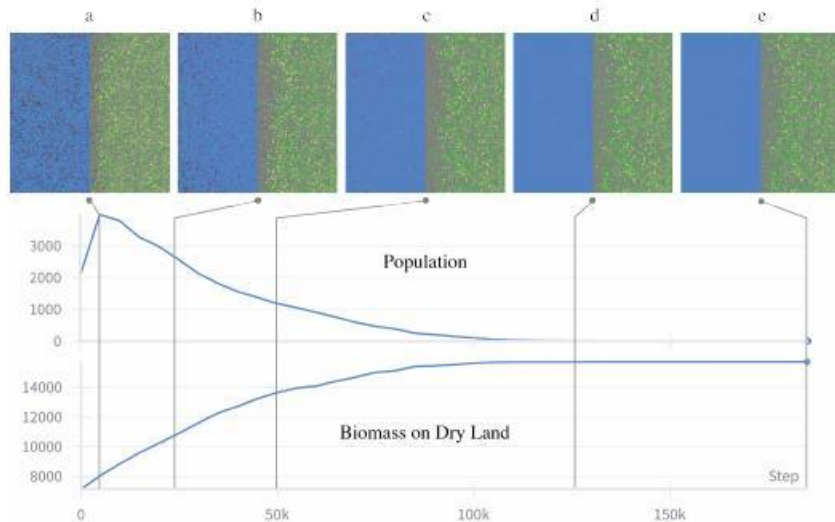


Figure 3: Representative trial with (**R**) agents in a 256×256 **Beach** environment. The top panel reports population size over time, and the bottom panel reports free biomass on dry land.

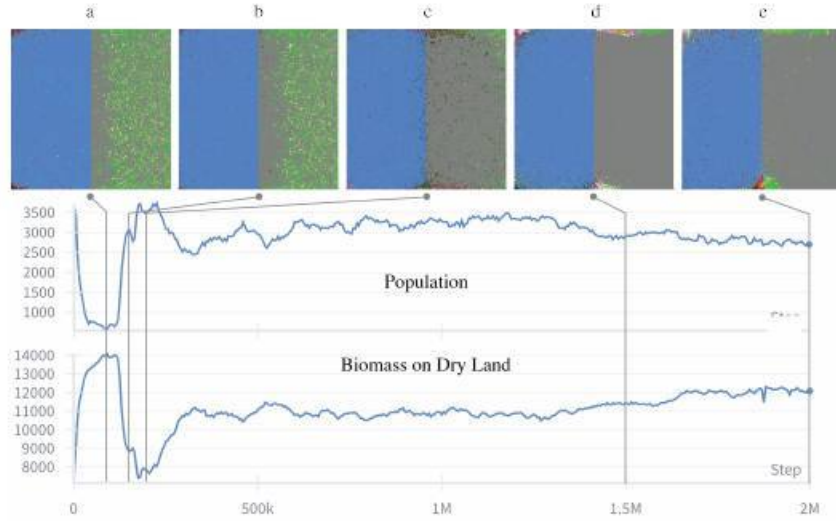


Figure 4: Representative trial with **(RC)** agents in a 256×256 **Beach** environment. The top panel reports population size over time, and the bottom panel reports free biomass on dry land.

Figure 3 illustrates the baseline **(R)** dynamics. After 5000 time steps (a), all agents initialized on dry land have died because they lack access to water, while agents initialized in water have rapidly expanded using the initially abundant aquatic resources. At this early stage, the population has not yet evolved policies that reliably seek resources. From the initial population peak, agents in the water occasionally move onto land (b). Because they lack a directional cue for returning to the ocean, many die inland and deposit their carried biomass there. By 50,000 time steps (c), this process has nearly doubled the amount of uneaten biomass on land. The resource transfer from water to land reduces the biomass available to sustain the aquatic population; by 125,000 steps (d), nearly all biomass has been transferred to land and the population is close to extinction. The final agent dies around 185,000 steps (e).

This trajectory shows that individual navigational failure can produce a population-level ecological transition. Deaths do not merely remove agents; they redistribute conserved biomass into regions that the surviving population cannot reliably exploit. The same redistribution also creates an adaptive opportunity: any lineage capable of entering land, consuming the accumulated biomass, and returning to water can access a large resource reservoir unavailable to the baseline population.

Figure 4 shows the corresponding behavior when agents receive an internal compass **(RC)**. The early dynamics are similar: the population initially shrinks and free biomass on land rapidly increases (a). With compass information, however, some agents can drift into the relative safety of the corners. Around 150,000 steps (b), agents begin to “mine” the biomass on land by moving between the shoreline and the interior, using directional information to return to water. By 200,000 steps (c), these lineages have cleared the beach. The population subsequently reorganizes into edge clusters (d) while remaining largely stable through 2,000,000 time steps (e).

We quantify this pattern by running **Beach** experiments at grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively. Each condition is evaluated with four random seeds. We record a mining event when free biomass on dry land drops by at least 10% from its historical maximum; we also record the onset time of the event, defined by the corresponding local maximum in land biomass, and the subsequent percentage drop. Table 1 summarizes mining and extinction events, and Table 3 in Appendix A reports seed-level details.

The summary shows a clear sensor and scale dependence. Mining occurs in all four **(RC)** seeds at 1024×1024 , 512×512 , and 256×256 , while the two smaller worlds show both fewer mining events and more extinctions. In contrast, **(R)** populations almost never mine successfully and go extinct in all four seeds at every size except 1024×1024 , where one seed exhibits a mining event and no seed

Table 1: Mining events in the **Beach** world in populations of agents with a compass (**RC**) compared to populations without a compass (**R**) across five world sizes and initial populations.  denotes the number of seeds that featured a mining event.  denotes the number of seeds in which the population went extinct before 2M steps. See Table 3 in Appendix A for detailed data from each seed.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

goes extinct before 2M steps. These results support the interpretation that compass information is a key enabling cue for long-distance biomass recovery in the **Beach** geometry, and that larger worlds make this behavior more reliable in the evaluated conditions.

To examine timing and extinction more systematically, we run additional (**R**) and (**RC**) seeds at 512×512 world size (32 seeds) and 128×128 world size (64 seeds). We use Kaplan-Meier estimators to model time to extinction and time to mining while treating runs without an event by 2M steps as right-censored. We also fit Cox proportional hazards models to compare event timing across world size and compass access. The appendix analysis shows that both larger world size and compass access reduce extinction hazard, while larger worlds and compass access accelerate mining emergence under the stated event definition. Full details are provided in Appendix B.

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains in populations of agents with a compass (**RC**) compared to those without a compass (**R**) at the 512 × 512 grid size.  denotes the number of seeds that featured a mining event.  denotes the number of seeds in which the population went extinct before 2M steps. See Table 4 in Appendix A for detailed data from each seed.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

We next test whether the compass effect persists beyond the **Beach** terrain by running **Island**, **Lake**, **Isthmus**, and **Channel** experiments at the 512 × 512 grid size with (**R**) and (**RC**) agents. Table 2 reports mining and extinction events, and Table 4 in Appendix A provides seed-level data. Across these terrains, compass-equipped populations generally exhibit earlier and more consistent mining as well as lower extinction rates. The effect is strongest in the **Island** and **Isthmus** terrains, where moving straight east or west reliably returns an agent to water; it is weaker in the **Lake** and **Channel** terrains, where no comparably simple directional rule exists. Baseline (**R**) agents do not reliably adapt to any of these geometries, though they struggle least on the island. Figure 12 in Appendix D visualizes mining in the **Island** terrain.

4.2 VISUAL FORAGING AND PREDATION

The second set of experiments evaluates whether agents can use vision to locate free resources and prey in the **Ocean** environment. We compare (**RCV+A**) agents, which have resource sensors, a compass, a 7 × 7 visual sensor, and attacks, against (**RC+A**) agents, which have resource and compass sensors with attacks but no vision.

Figure 5 plots population, biomass utilization, move actions, and eat actions for 512 × 512 and 128 × 128 grids. Additional grid sizes appear in Figure 9 in Appendix C and Table 5 in Appendix

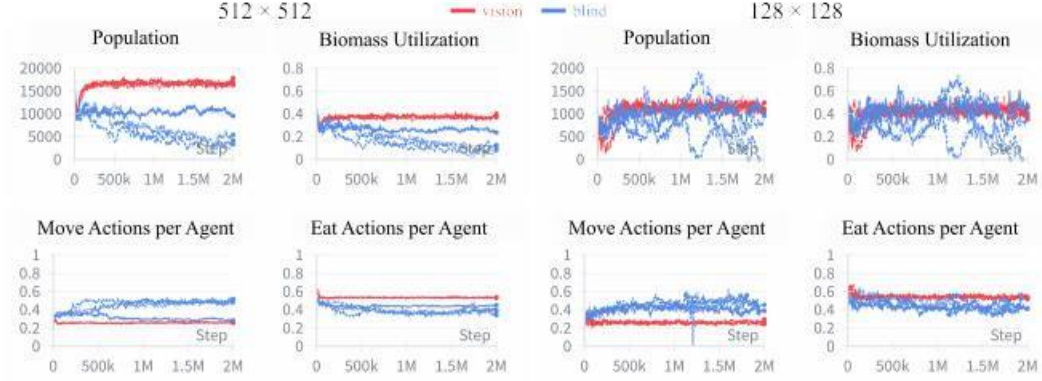


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). Vision-enabled populations are larger and more stable, with greater biomass utilization produced by less movement and more frequent eating. The larger scale shows lower variance across runs, consistent with a stronger foraging advantage from vision in these environments. See Figure 9 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

A summarizes the corresponding statistics. Biomass utilization is the percentage of total environmental biomass currently contained inside agents. Across the plotted conditions, visual agents spend less time moving and more time eating, sustain larger populations, and maintain greater biomass utilization. These patterns indicate that visual sensing supports more efficient foraging. Across scales, between-run variance decreases substantially in larger environments, and the vision runs exhibit more stable temporal dynamics than the corresponding blind-agent runs.

Figure 6 reports attacks per agent and homicides per attack for the same runs. Visual agents attack relatively infrequently while maintaining a successful kill rate of approximately 75%. Blind agents in large grids sometimes also achieve high kill rates, but their dynamics are much less stable. The causal interpretation is limited by confounding population configurations: visual inspection (Figure 13 in Appendix D) indicates that high blind-agent kill rates can occur when dense clusters form in one location, whereas such clusters rarely appear in visual-agent runs. This pattern is consistent with the possibility that vision supports both targeted attacks and avoidance, but the present experiments do not disentangle these mechanisms. In smaller environments, the difference between visual and blind agents is less pronounced, although visual-agent populations remain substantially more stable.

4.3 ABLATIONS

Appendix F evaluates the sensitivity of the reported phenomena to simulator configuration. The ablations vary initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8). Appendix F.3 additionally tests whether the more stable dynamics in larger environments can be explained solely by self-averaging over more agents.

5 CONCLUSION

This study demonstrates that large-scale ecological simulation is a productive experimental paradigm for investigating how adaptive artificial behaviors can arise from survival, reproduction, and competition rather than from externally specified task rewards. Across terrains, sensor configurations, and world sizes, we find that environmental scale and sensory richness materially affect both population stability and the kinds of behaviors that emerge, including compass-enabled long-range biomass recovery and vision-enabled foraging and predation in ocean environments. The results should be interpreted as evidence for structured ecological adaptation in the studied simulator, not

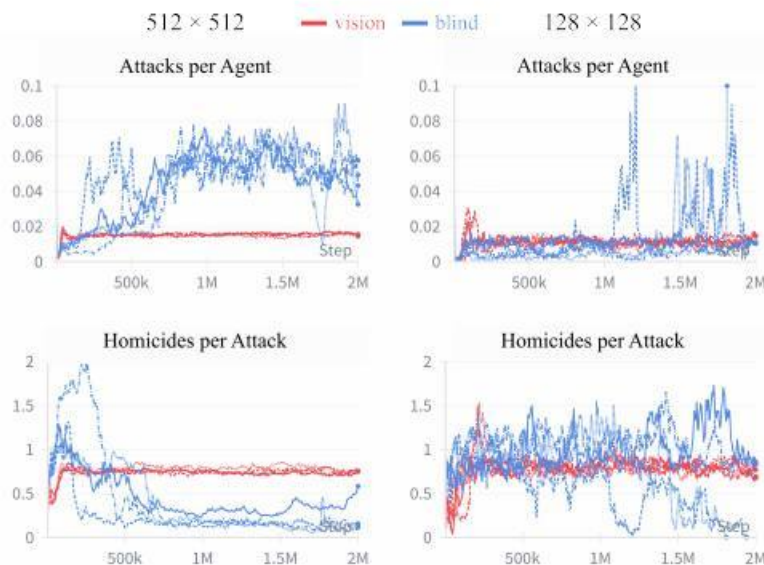


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). Vision-enabled agents attack less frequently but with much higher precision, suggesting specialized predation behavior within these conditions. The separation is more pronounced at larger scale. See Figure [10] in Appendix [C] for a comparison of grid sizes 1024×1024 and 256×256 .

as a demonstration of unrestricted general intelligence or universal open-endedness. Within that scope, they show that modern accelerator hardware can support large evolving populations with individualized neural policies, making ecology a practical substrate for controlled studies of behavioral emergence. We hope these findings encourage more systematic exploration of which environmental mechanisms, sensory channels, and scaling regimes are necessary for increasingly rich artificial ecologies.

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A ADDITIONAL DATA

Table 3: Mining events in the **Beach** world in populations of agents with a compass (**RC**) compared to populations without a compass (**R**) across five scales.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			55
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k	85.3	675	
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓	10k	78.	n/a
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			135
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4	40k	81.7	3/4
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			45k
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			40
2	x			n/a	x			1070k	x			165k	x			55k	x			25
3	x			n/a	x			280k	x			125k	x			90k	x			40
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			4/4

Table 4: Mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** among (RC) agents compared to (R) agents. Grid size is 512×512 with an initial population of 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging data for Vision and Blind agents in the **Ocean** world at different scales. denotes population size. denotes biomass utilization, the percentage of total biomass in the system that is currently inside the agents' stomachs. denotes move actions per agent, the fraction of time an agent took a move action, averaged across agents. Similarly, denotes the proportion of agents' actions that took an eat action. Reported values of , , , and are averages of the metric over all steps for the seed.

	1024x1024				512x512				256x256				128x128			
	32768				8192				2048				512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation data for Vision and Blind agents in the **Ocean** world at different scales.  denotes attacks per agent, the fraction of time an agent took the attack action, averaged across all agents. The  denotes homicides per attack, the number of agents killed per attack action. Reported values of  and  are averages over all steps for the seed. Note the outlier of 0.562 for  of seed 3 in the 128×128 (**RC+A**) run. This seed went extinct before 2M steps, resulting in a spike in the attacks per agent metric due to dwindling population size.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We compare rates of extinction and mining emergence dependent on world size and agent access to global direction via the compass sensor. Extinction is defined as the population reaching zero individuals, while mining behavior is defined as it was in earlier experiments (a 10% drop in free biomass on land).

Runs are grouped based on both world size and compass access. We simulated 64 runs each of 128×128 with no compass and 128×128 with compass, and 32 runs each of 512×512 with no compass and 512×512 with compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution (treating runs that did not go extinct within 2M steps as right-censored). In Figures 7 and 8 we observe that worlds tend to survive longer and produce mining more quickly if they are large or if agents have compass information.

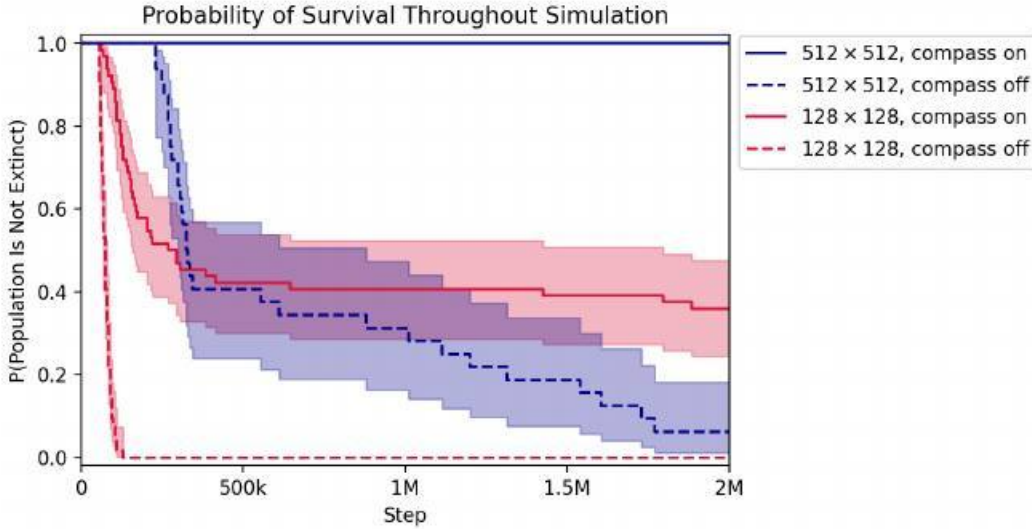


Figure 7: Probability the population has not gone extinct given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan–Meier estimator for each group of runs (e.g. 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

We compare emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7). Runs that did not experience an emergence or extinction event by the end of 2M time steps were treated as censored. For extinction, both larger world size and compass use substantially reduce the hazard; the extinction rate in 128×128 is about 25 times greater than the corresponding rate in 512×512 worlds ($HR=0.04$). Enabling the compass produces an effect of a similar magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining behavior much more rapidly ($HR=6.07$), and compass use accelerates emergence dramatically ($HR=34.59$). The models show good discriminative ability between groups (concordance 0.80 for extinction, 0.85 for mining). We find that larger worlds can sustain larger populations, reducing the chance of hitting the absorbing zero boundary while simultaneously exploring a larger phenotype space due to greater quantities of mutation as compared to smaller worlds.

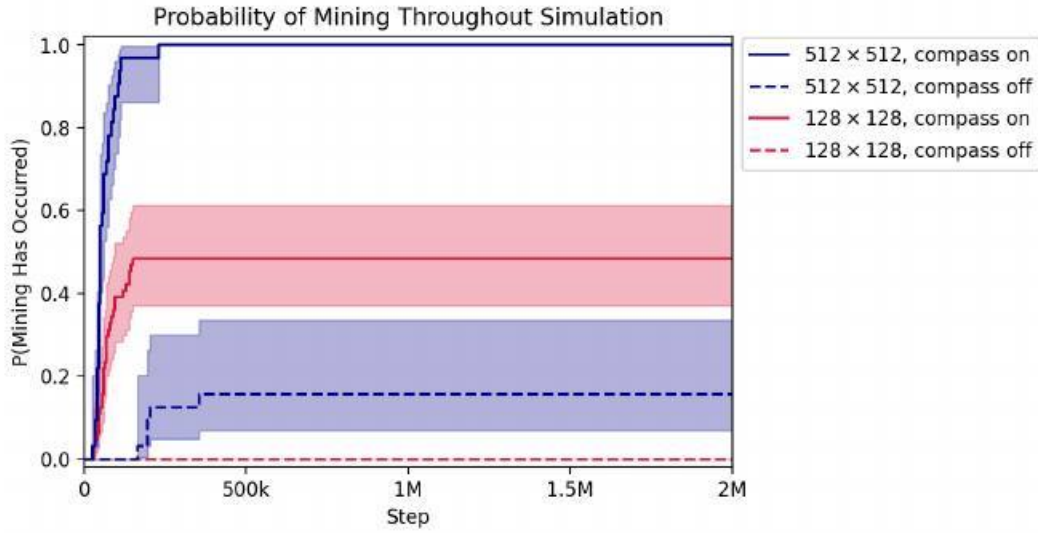


Figure 8: Probability mining has evolved within the population given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan-Meier estimator for each group of runs (e.g. 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events with a 128×128 , compass off baseline. Hazard ratios (HR) less than 1 indicate reduced hazard (longer time to event); for example, extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the rate of 128×128 . There are 192 total runs: 32 seeds with compass on and 32 seeds with compass off in the 512×512 world, and 64 seeds with compass on and 64 seeds with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

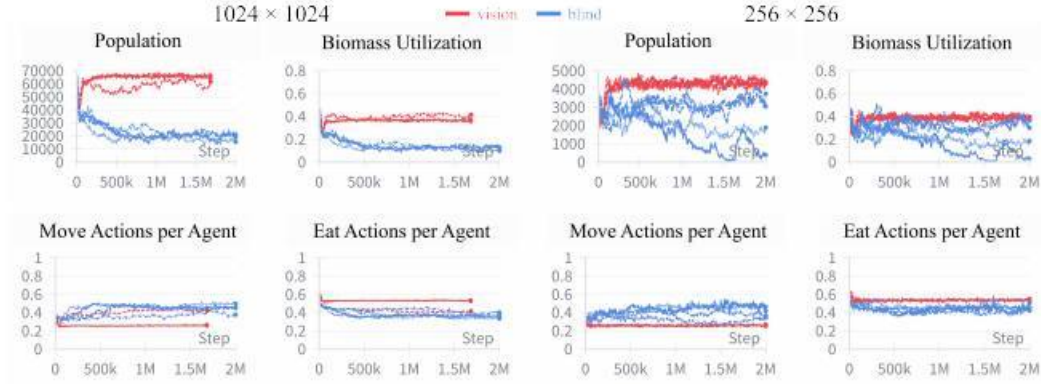


Figure 9: Foraging behaviors in the **Ocean** world at different scales. Enabling vision in agents (red) results in more stable populations and greater biomass utilization as a result of more frequent and effective eating and less movement. At larger scales, these dynamics are more stable with less variance across runs. Note that due to limitations on GPU hours, the vision seeds at the 1024×1024 scale ran for 1.7M instead of 2M steps.

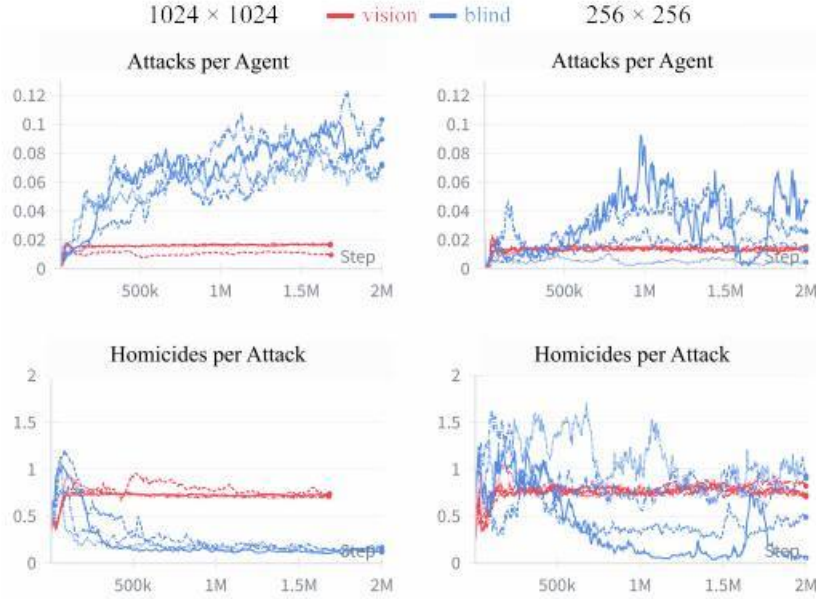


Figure 10: Predation behaviors in the **Ocean** world at different scales. With vision (red), agents attack less frequently but with much higher precision, suggesting the emergence of specialized predation behaviors in addition to foraging capabilities (Figure 9). This dynamic is again more pronounced at larger scales, with more variance in the behaviors of blind agents (blue) in smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

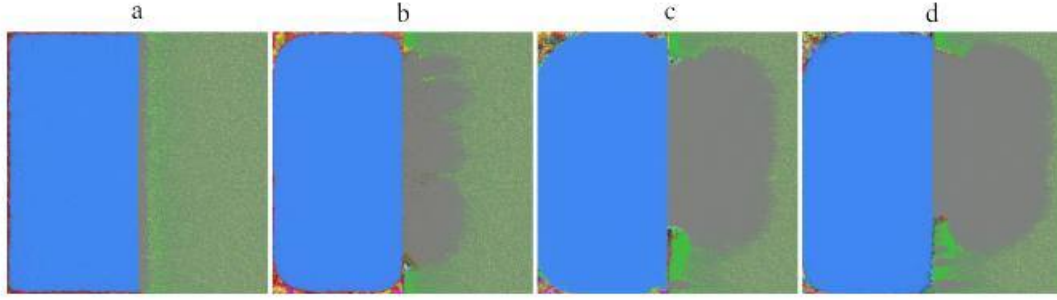


Figure 11: Larger scale environments induce more varied and complex ecologies that give rise to unique visual patterns. In this 1024×1024 **Beach** world, (RC) agents initially died at the shore (a), transferring biomass to dry land. They then developed mining behavior, clearing biomass from the beach (b). When biomass re-accumulated on the ends of the beach (c), subpopulations of agents re-developed mining behavior, traveling inland on these mini biomass beaches to extract the resources (d).

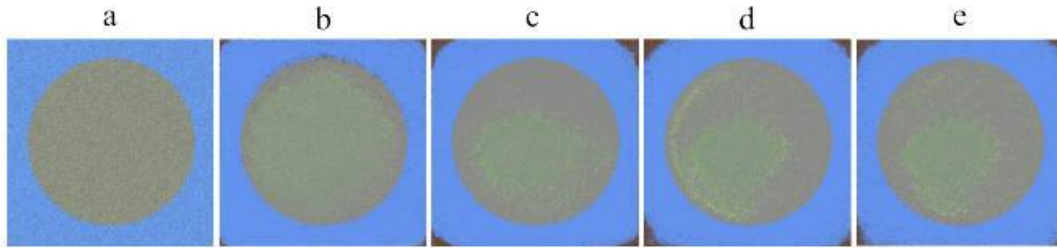


Figure 12: Mining behavior emerges in varied terrains. In this 512×512 **Island** world with (RC) agents, different regions of the shore are mined throughout the run (b) (c). Biomass is transferred back to parts of the island once mining becomes difficult and agents die before they can return to water (d). Agents adapt to re-mine this wall of accumulated biomass, clearing it from shore (e).

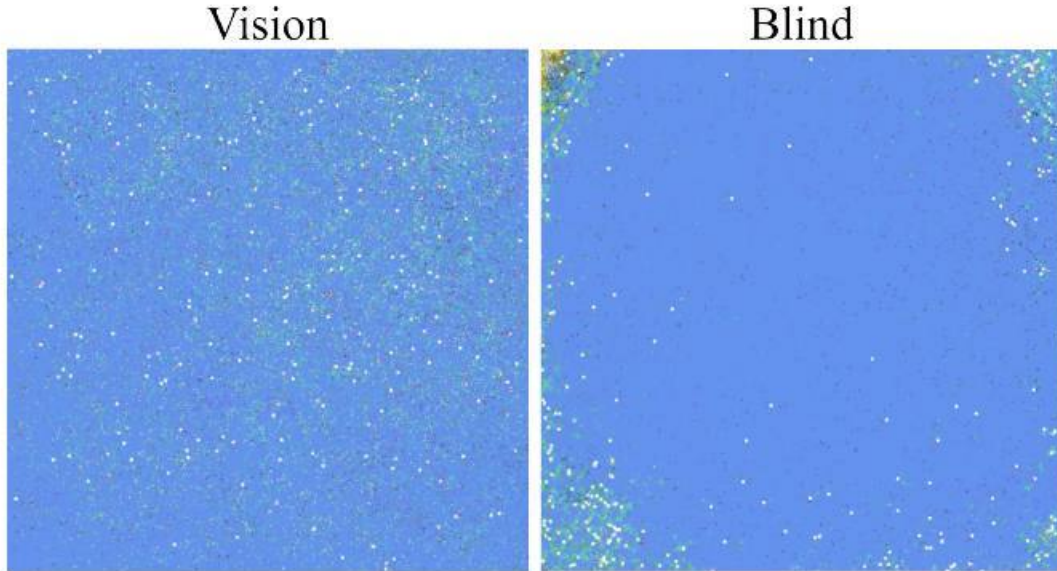


Figure 13: Representative runs in a 512×512 **Ocean** world where agents have attacking enabled: vision (RCV+A) agents on the left and blind (RC+A) agents on the right. The white squares indicate an agent is attacking at that location. With vision, agents adapt to locate free resources throughout the ocean world and prey on other agents more effectively (see Figures 5 and 6). Blind agents explore the ocean for resources to a much lesser extent, instead adapting policies for clustering in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We report details on simulation design, runtime, and experiment reproducibility.

Simulator design. We conduct all experiments using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. Our simulator computes every part of the environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation allows the same environment implementation to be reused with different evolutionary algorithms, reinforcement learning methods, or other population update rules. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU given in Table 8.

A key feature of the simulator is its built-in tooling, including a measurement API for computing and recording aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters at each step, which we use directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports, and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating qualitative phenomena such as long-range resource mining and predation patterns that are not obvious from scalar metrics alone.

From a high-performance computing perspective, our simulator fills a gap relative to existing simulators. General-purpose agent-based frameworks such as MASON (Luke et al. 2005), FLAME-GPU (Richmond et al. 2024), or Griddly (Bamford et al. 2020) offer high configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments emphasize fixed numbers of agents and task-specific reward structures. We combine GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination yields a state-of-the-art platform for scaling ecological experiments like those in this paper and a useful foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 shows runtime data across different grid sizes and maximum populations of agents. Each run was conducted on a single H100 GPU for a maximum of 2M steps. Note that while this table shows run-times for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), the runtime varies with the size of the agent networks, the available sensors, and the level of logging.

Our largest runs at the 1024×1024 grid size take ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on one H100, with memory usage of roughly 60GB. **Ocean** runs are slightly slower due to the enabled agent attack action and vision sensing.

Table 8: Runtime data in the **Beach** and **Ocean** worlds across different grid sizes and maximum populations of agents. The (RC) agents each have a 22,465 parameter neural network. The (RCV + A) agents each have a 22,529 parameter neural network. Simulation steps per second (hz) and total extrapolated wall-clock runtime in hours for 2M steps are reported. Effective runtime is less for seeds in which populations go extinct before 2M steps.

Environment	1024×1024		512×512		256×256		128×128		64×64	
	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

Reproducibility. Due to the large and randomized nature of our environments, there can be substantial differences in the precise outcomes of each experiment from run to run. As a result, we report the results of four different random seeds for each experiment. Unfortunately, due to nondeterminism in some low-level GPU operations, we find that repeating an experiment with the same random seed does not reliably produce the same results. In particular, modern GPUs may execute parallel floating-point operations such as summations in different orders across runs and fuse operations in ways that change rounding behavior. Very small differences arising from finite-precision arithmetic can accumulate over many steps into noticeable drift in the simulation trajectory. The use of 16-bit weights can coarsen these rounding effects, but is not the primary source. Despite differences across runs, clear high-level patterns emerge when the results of multiple seeds are taken in aggregate.

F ABLATIONS

We present a series of ablations investigating the effects of specific hyperparameters and robustness of results under different environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate the effect of initial population size in a 256×256 **Beach** world on the consistency of mining behavior and population stability. Results reported are for **(RC)** agent populations. Agents without compass **(R)** had 0% mining rate and 100% extinction rate across all initial population sizes except 8192, the highest tested, with 25% mining and 75% extinction frequency. Note that mining and extinction frequencies need not add to 100%: it can be the case that a population develops mining but then goes extinct, for instance, or that a population lasts for the full 2M steps of our experiments without developing the mining behavior.

In Figure 14 we find that the initial population size has no significant effect on mining consistency or extinction rates: populations starting with only 128 agents evolve mining behaviors as frequently as populations starting with 8192 agents, and extinction rates are largely stable across initial population size.

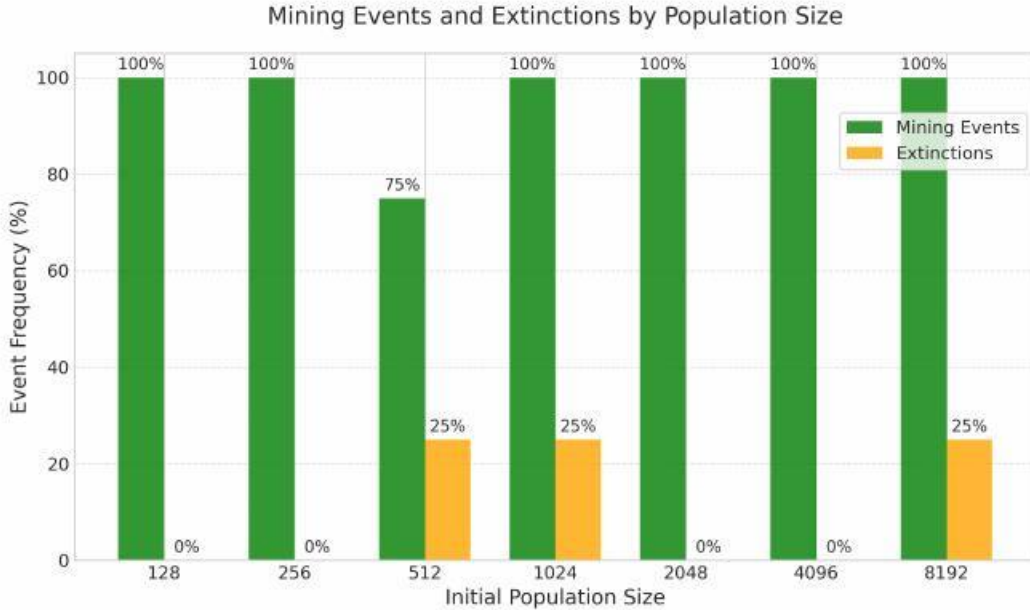


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world for different initial sizes of **(RC)** agent populations. Frequencies are computed across four random seeds for each initial population size; i.e. 25% Extinction Event Frequency means that in 1/4 seeds, the population went to 0 before 2M simulation steps.

Figure 15 provides some insight into why so little effect is visible. Regardless of the size of the initial population, each run rapidly converges to a similar population size in the first 50,000 steps of simulation. This suggests that the environment implicitly specifies a carrying capacity based on the fixed resources available in the system: populations beginning below the carrying capacity grow to reach it, and populations that are too large are constrained to the capacity. As a result, the effective population size is similar after an early correction of the initial population size, leaving mining and extinction frequencies unaffected. Importantly, however, larger initial population sizes contribute somewhat to the total resources in the system because agents carry biomass that is deposited upon death; as a result, larger initial population sizes are able to reach higher maximum populations on average over the course of simulation (Table 9), but the effect is not significant enough to impact mining or extinction frequency for this setting. It is worth noting that although the population size converges early on, it does not stay consistent throughout the run. Later in the simulation, the populations between different runs can vary substantially, as the ecological impacts of agents evolving different abilities at different times changes the later resource distributions.

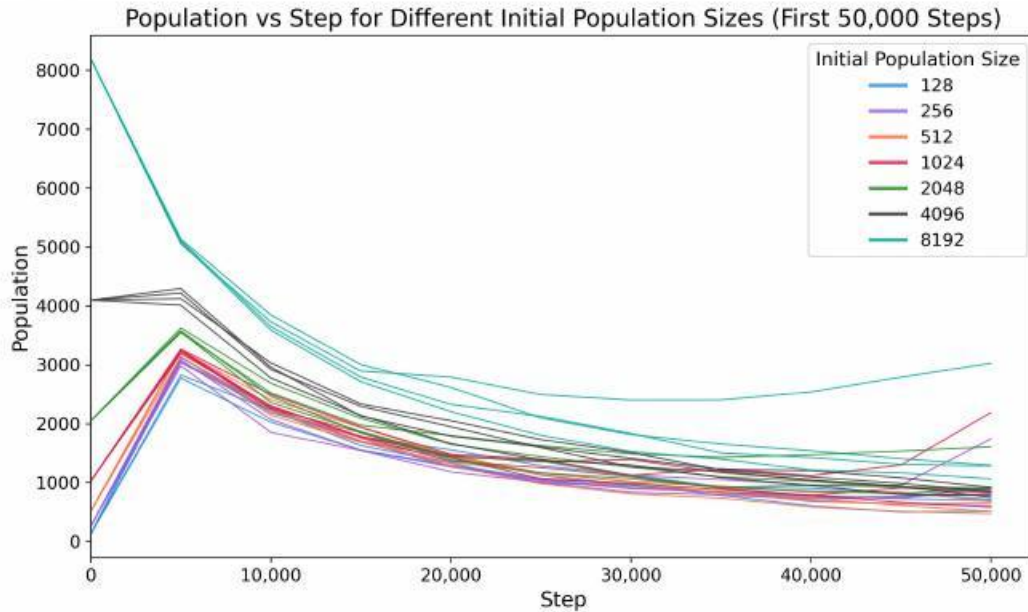


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with **(RC)** agents, with four seeds per initial population size. The environment supports an implicit carrying capacity based on resources available in the system, and all initial population sizes approach this value early in the run. Larger initial population sizes contribute to the total resources in the system due to biomass deposited on agent death, raising the carrying capacity somewhat.

F.2 VARYING RESOURCE RULES

We evaluate the effect of two resource hyperparameters, biomass photosynthesis rate and initial biomass site density, on mining and extinction frequency. The setting is equivalent to that in Appendix F.1. The initial population size is fixed at 2048 for this study, and agents once again have compass **(RC)**.

The biomass photosynthesis rate controls the rate at which landscape biomass creates energy via photosynthesis, providing resources for agents over time. The initial biomass site density parameter determines the fraction of grid cells that have biomass at the start of simulation. All experiments in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

While varying initial population size did not affect mining and extinction consistency (Figure 14), changing resource rules has a significant effect as demonstrated in Figures 16 and 17.

Table 9: The maximum population size reached after step 0 for different starting population sizes, with averages and standard deviations across four random seeds. Larger initial population sizes reach slightly higher maximum populations with greater variance across seeds than for lower initial population sizes. However, the max population size reached is largely robust to the initial population size: an initial population size (128) of only 1.6% of an initial size of 8192 reaches 78.9% of the max population reached by the latter (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 $\pm$ 217
256	3056 $\pm$ 61
512	3451 $\pm$ 437
1024	3613 $\pm$ 381
2048	3786 $\pm$ 431
4096	3689 $\pm$ 437
8192	4026 $\pm$ 642

Intuitively, larger biomass photosynthesis rates or initial biomass site densities directly boost the effective carrying capacity of the environment by increasing the resources available in the system. As a result, larger populations are supported throughout the run, increasing the likelihood of mutations favoring mining behavior while lowering the chance of extinction. Our data support this intuition: on average (omitting seeds that went extinct), runs with initial biomass site density 1/16 reached a max population of 1608 after step 0, while reaching a mean max population of 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

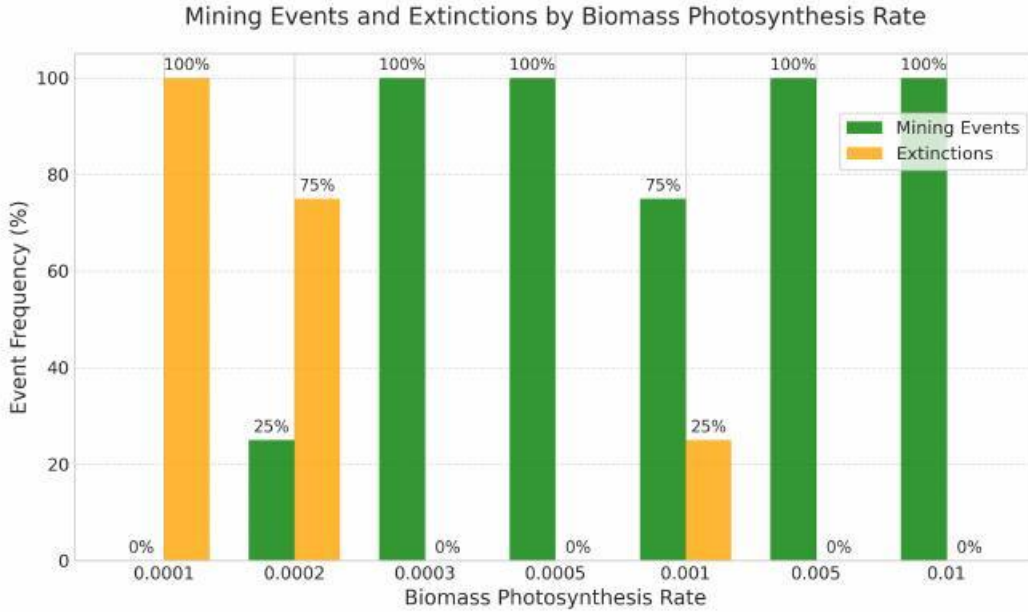


Figure 16: Frequency of mining and extinction events in a 256 $\times$ 256 **Beach** world for different settings of the biomass photosynthesis rate resource parameter. Frequencies are computed across four random seeds for each setting. Mining consistently emerges with biomass photosynthesis rates of 0.0003 and greater.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

A concern is that the increased stability of mining behavior in large environments (Table 1) might be explained purely by self-averaging rather than by qualitatively different dynamics. By self-

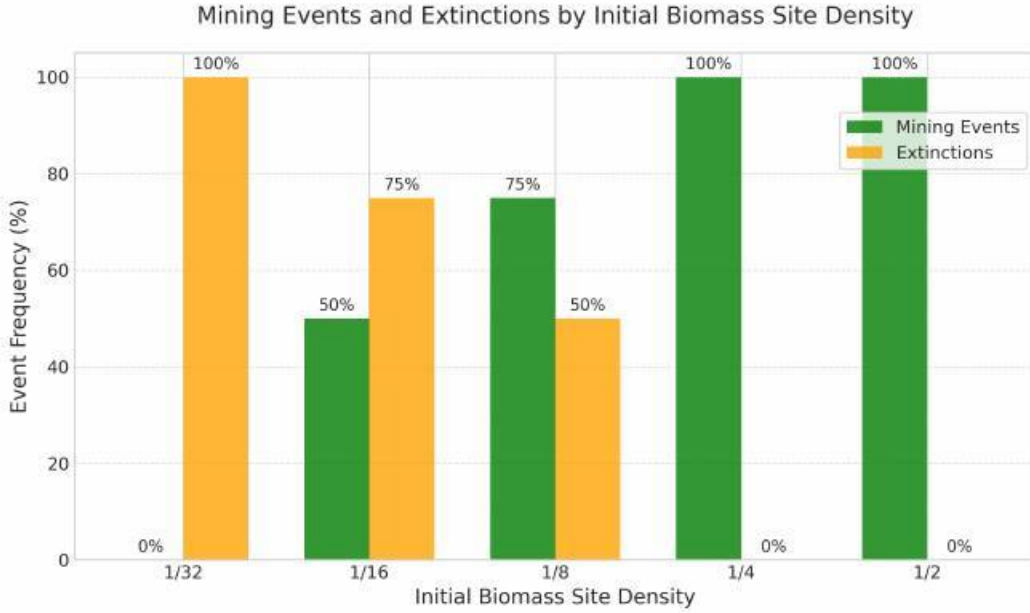


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world for different settings of the initial biomass site density resource parameter. Frequencies are computed across four random seeds for each setting. Greater initial biomass site densities result in more consistent discovery of mining and lower chance of extinction.

averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, such that a single large system behaves like the average of smaller systems of the same total size. Under this view, a 512×512 **Beach** world might simply be acting like many loosely coupled 128×128 worlds run in parallel. If that were the case, then our scaling results could in principle be reproduced by running enough small worlds and averaging over random seeds.

To test this hypothesis, we run an experiment holding the total “sample size” roughly fixed while changing how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat this protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

If large environments were acting as self-averages of many smaller ones, we would expect mining and extinction rates to be roughly invariant to how we distribute grid cells and agents across runs: 4 large worlds or 64 small ones would exhibit similar probabilities of discovering mining. Instead, we observe a strong dependence on world size. For (**RC**) agents, mining emerges in all runs with no extinctions at 512×512 , remains common but not guaranteed at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before reaching 2M steps. In contrast, (**R**) agents never develop mining and go extinct at all scales, confirming that the compass remains a key enabler of long-range behavior.

Table 10: Mining and extinction events when trading off world size against the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , 128×128 with initial populations scaled proportionally (8192, 2048, 512 agents, respectively). For each size, we allocate the same total number of grid cells and initial agents across all runs by running 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged (🔍) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass (RC) and without a compass (R).

	512 × 512 8192		256 × 256 2048		128 × 128 512	
	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results are inconsistent with a self-averaging explanation. On the contrary, larger environments appear to enable and stabilize mining behavior in ways that cannot be explained solely by averaging over more small simulations.

Even when matching the total number of cells, agents, and simulation steps across conditions, concentrating resources into a few large worlds produces a regime in which mining is consistent and extinctions are rare, while dispersing the same resources across many small worlds gives populations that frequently fail to discover mining and are more likely to collapse. This suggests that increasing environmental scale is not simply smoothing out stochastic noise, but is altering the underlying ecological dynamics, for example by supporting larger, more spatially distributed populations and more persistent diversity in evolving policies.

F.4 MUTATION RATE SWEEP

We test the sensitivity of the emergence of mining to the mutation rate hyperparameter, which controls the scale of Gaussian noise added to a child agent’s policy upon birth. Intuitively, the mutation rate directly impacts how different a child agent is from its parent. We once again run experiments in a 256×256 **Beach** world with an initial population of 2048 (RC) agents for a maximum of 2M steps.

In Figure 18 we show that between mutation rates of 6×10^{-3} and 7.875×10^{-2} , mining emerges in at least 3 of the 4 seeds run, indicating that our results in Section 4.1 with mutation rate 3×10^{-2} are robust within this interval. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, leading to extinction. Conversely, large mutation rates (e.g., 3×10^{-1}) undermine population stability, consistently leading to extinction with no mining adaptation.

F.5 NETWORK SIZE

We study the effect of the size and architecture of the agent policy network on mining and extinction frequency, as well as the impact of environmental scale on this effect. We investigate 256×256 and 512×512 **Beach** worlds with initial population sizes of 2048 and 8192 (RC) agents, respectively (scaling by four with the number of grid cells). We ran baselines with (R) agents for each experiment, again finding rare to no mining events and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, yielding a network of 22465 parameters per agent (Table 11). In Figure 19 we show that in the 256×256 **Beach**, halving channels to 32 (resulting in 9185 total parameters) or doubling to 128 (giving 61313 parameters) with number of layers fixed has minimal to no effect on mining and extinction frequency. Meanwhile, changing network depth has a significant effect: all populations of agents with just 1 layer and 32 channels went extinct, and 50% of populations of agents with 3 layers and 128 channels went extinct.

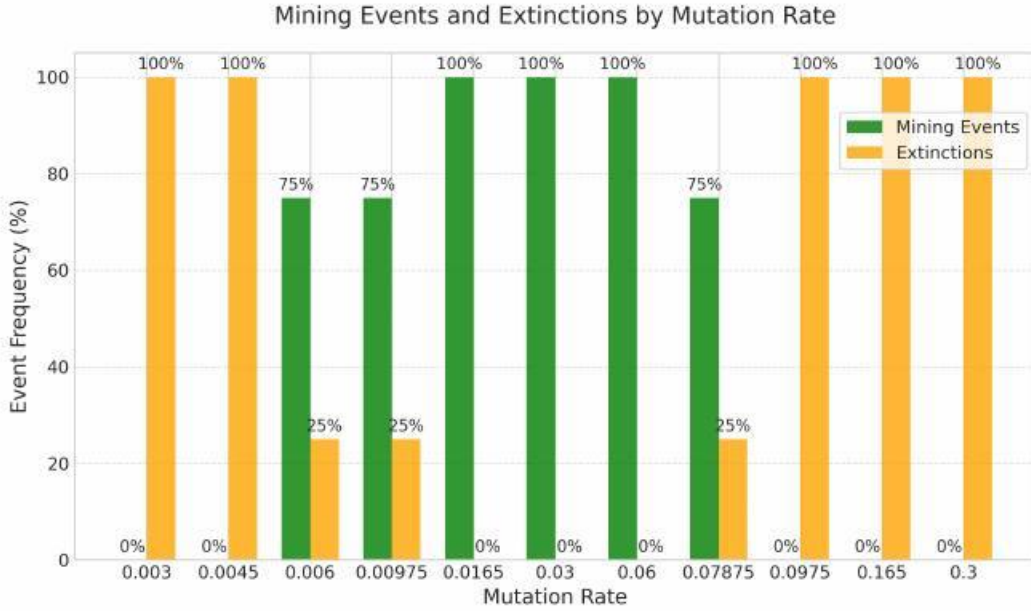


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world for different mutation rates. Frequencies are computed across four random seeds for each setting.

Crucially, scaling the environment to the 512×512 grid size produced significantly different dynamics for networks of varying depth (Figure 20). In particular, in the larger world, agent populations with 3 layer, 128 channel networks consistently mined and survived through 2M steps. Surprisingly, agent populations with the small 1 layer, 32 channel networks mined consistently as well. We hypothesize that it is more challenging for both smaller and larger networks to discover new behaviors for different reasons. When the network is smaller, the reduced capacity makes it hard to find good solutions. When the network is larger, the increased size of the search space means it takes longer to find promising new behavior. Regardless of the reasons, the fact that these networks fail so often in small environments but succeed in larger ones further illustrates the stability that comes from increasing scale. The larger environment can support larger populations, which allows for more exploration in policy space. Simultaneously, it takes longer for locally successful mutants to propagate across a large environment, which promotes diversity and again ensures greater exploration. For more discussion of these effects see (Mitchell et al. 2006).

Table 11: Total parameters in (RC) and (R) agent policies with different network architectures.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

F.6 VARYING VISION RANGE

We assess the impact of different vision ranges and viewing configurations on predation and foraging behaviors in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and 9×9 . For each range, we evaluate two viewing configurations shown in Figure 21 a) Around: agents can see grid cells all around them b) Front: agents cannot see cells behind them.

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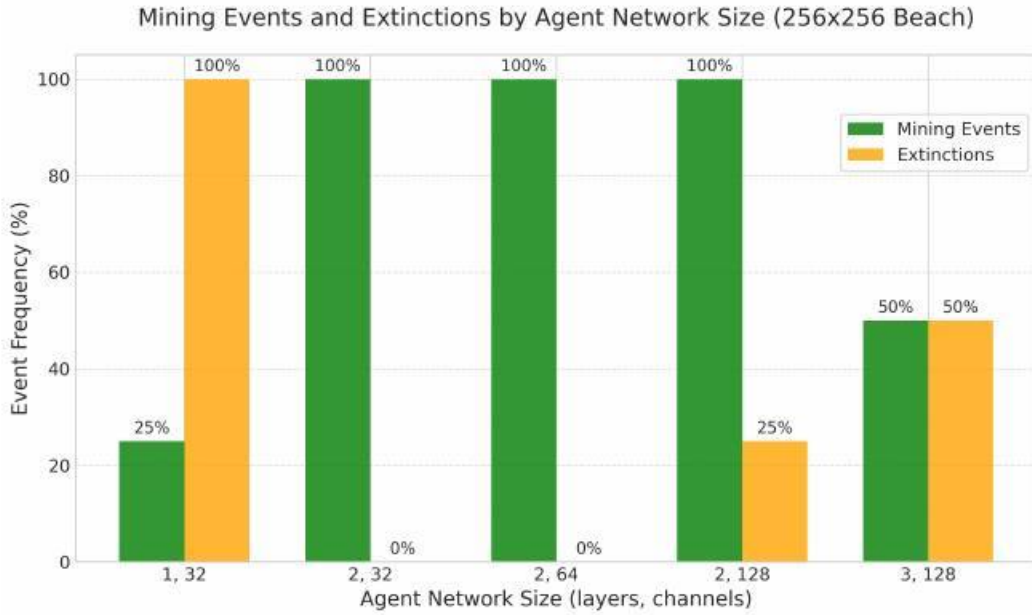


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

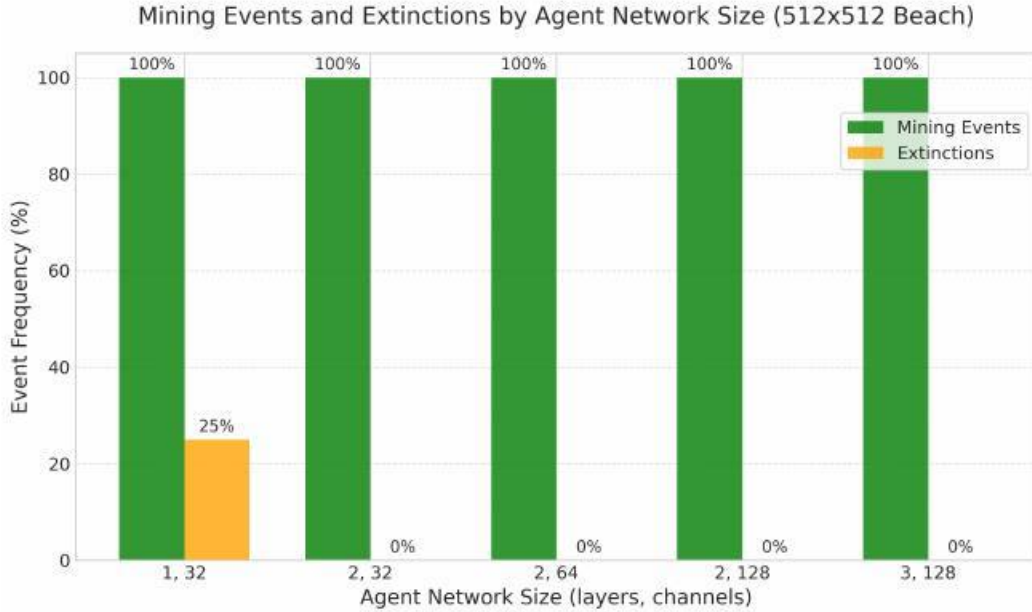


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

The total number of cells in the agent’s vision range is equal for the two configurations. The main experiments with vision in Section 4.2 were run with the 7×7 Around setting.

These results indicate that a smaller 3×3 vision range results in behavior that is largely similar to no vision at all (Blind), while the results at 9×9 appear similar to 7×7 . This indicates that there may be diminishing returns of larger vision ranges in the current environmental configuration. This is intuitive given the rest of the environmental configuration; since the agents have no memory, it is

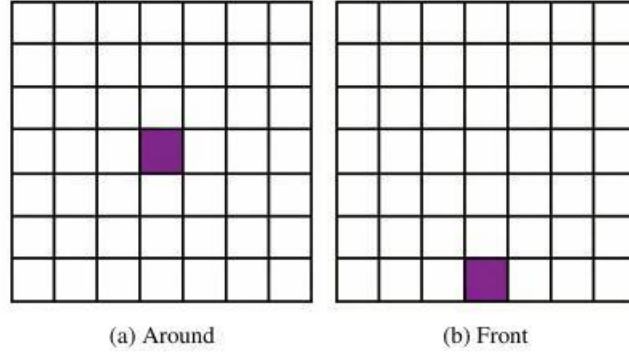


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell represents an agent, facing the top of the page.

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) with varying vision ranges and viewing configurations. Refer to Tables 5 and 6 for definitions of the metrics represented by each icon. For each seed, the value for each metric was averaged across all steps starting at step 500k to allow them to stabilize. Reported values are the mean across the four seeds and the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

unlikely that visual features at the periphery of a 9×9 window could meaningfully alter the optimal action relative to a 7×7 window consistently enough to provide a strong learning signal.

The data also suggest that at the 7×7 and 9×9 vision ranges, the Around configuration is slightly more favorable in terms of population stability and effectiveness of attack and eat actions, resulting in higher average population , biomass utilization , and homicides per attack  than the Front configuration. We hypothesize that this is due to diminishing returns of being able to see more distant pixels; in the Around configuration, more pixels in the agent’s vision range are nearby, likely containing more relevant information for near-term actions than distant cells visible with the Front configuration. At the 3×3 vision range, however, the trend is reversed, with the Front configuration being more favorable than Around; this indicates that in this environment, the marginal benefit of seeing one more pixel ahead through the 3×3 Front configuration outweighs that of seeing one pixel behind in the 3×3 Around configuration, which gives most similar behavior to the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study the effect of reducing agent attack strength on emergent behaviors and population dynamics in the 256×256 **Ocean** world. We evaluate the following attack strengths: 1) Full Attack: deals 10 HP of damage, instantly killing any agent in the attack radius 2) Half Attack: deals 5 HP of damage 3) Quarter Attack: deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 were run with the Full Attack setting.

We find that reducing the attack strength from Full Attack to Half Attack significantly reduces the effectiveness and frequency of attacks: attacks per agent and homicides per attack drop considerably (Figure 22). As a result, populations with lower attack strength grow larger, with agents moving and eating more frequently, resulting in greater biomass utilization. Further reducing strength to

the Quarter Attack setting yields diminishing returns in terms of further reduction in violence and facilitation of additional population growth. Intuitively, when attacks do less damage, a homicide requires more attacks per agent, reducing the lethality of each attack. However when the lethality is reduced, attacking behavior becomes less useful and therefore less likely to be selected for in the agent population. Agents with lower attack strengths therefore do not become as skillful at attacking and spend more time foraging for food.

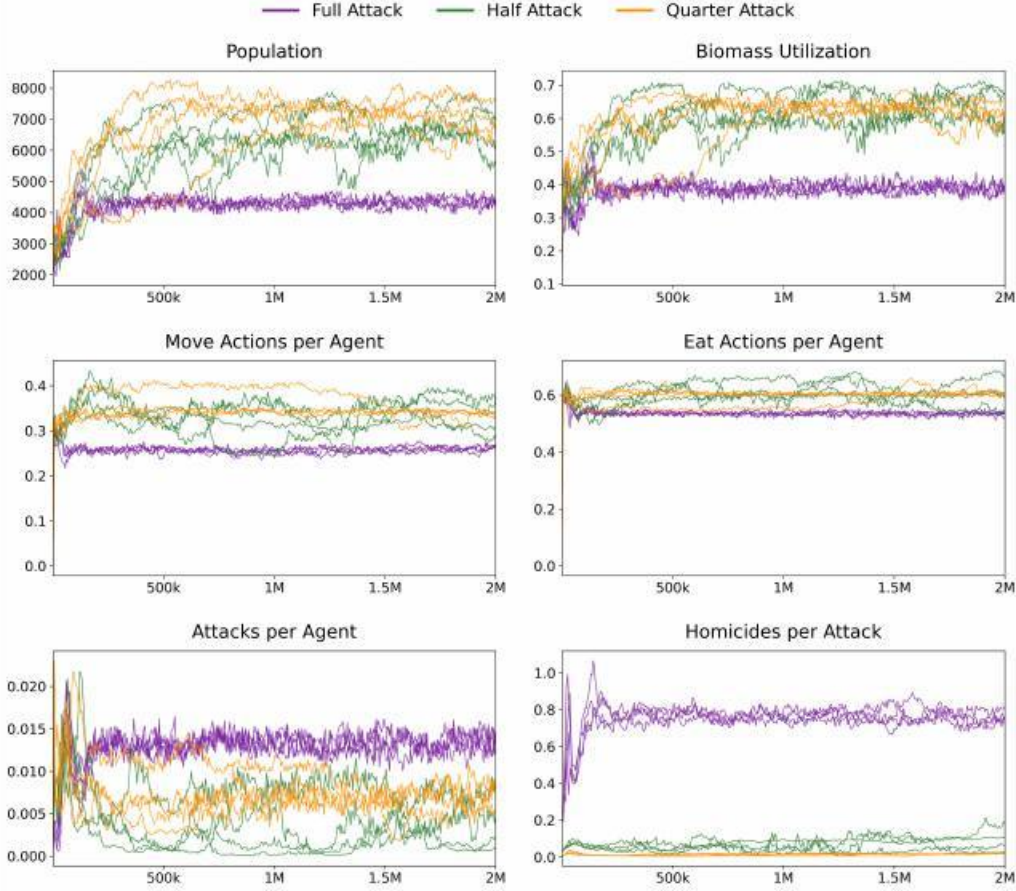


Figure 22: Population and action frequency data over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted per attack strength setting.

F.8 VARYING AGENT HP

We investigate the effect of reducing the starting HP of child agents and lowering the healing rate to apply pressure for agents to ‘grow’ over their lifetime in order to survive. The healing rate parameter determines how much HP an agent can regenerate per healing step, which converts available energy to HP. Agents can influence healing only by maintaining enough energy.

We study two settings: 1) Full HP: child HP starts at 5 and healing rate is 1 HP per 0.1 energy 2) Low HP: child HP starts at 1, with healing rate 0.5. Max agent HP is 10 in all settings. The main experiments (Section 4) were run with the Full HP setting.

Across the two settings, there is no significant difference in homicides per attack, eat action frequency, or biomass utilization (Figure 23). In the Full HP setting, populations grow slightly larger, while in the Low HP setting, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must focus on foraging early on in their lifetime in order to grow to full size, which would explain this difference.

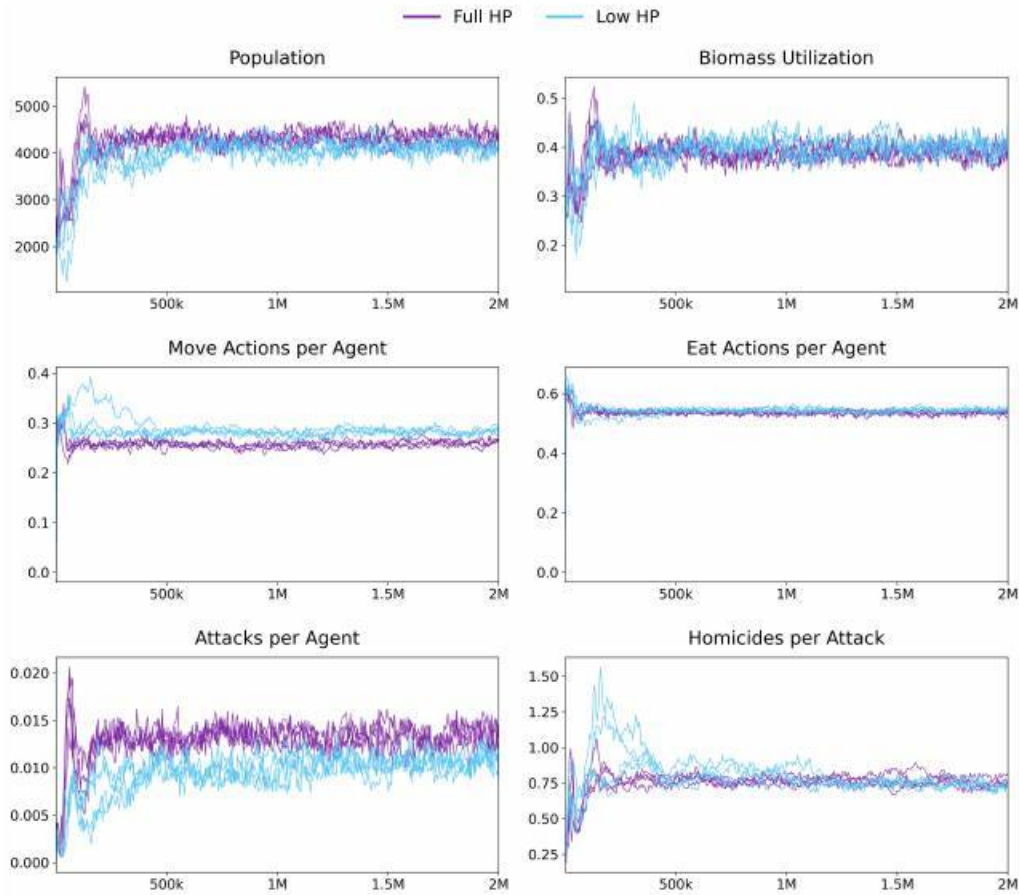


Figure 23: Population and action frequency data over 2M steps in the 256x256 Ocean world for (RCV+A) agents across Full HP and Low HP settings. Four seeds are plotted per HP setting.